Aadhya Anand

aadhya.anand@uwaterloo.ca | linkedin.com/in/aadhya-anand | aanand.uwce.ca

Technical Skills

Languages: Python, Java, C/C++, JavaScript, PHP, SQL (Postgres), SOQL, HTML/CSS, R
Frameworks & Libraries: React, Flask, pandas, NumPy, Matplotlib, Scikit-learn, UMAP
Developer Tools: Git, VS Code, PyCharm, IntelliJ, Visual Studio, PHPmyAdmin
Platforms & CRM: Salesforce, WordPress
Other: n8n, Altium Designer

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering — *Sep. 2025 – Apr. 2030*

- Relevant Coursework: Digital Circuits (ECE124), Linear Circuits (ECE140), Accelerated Numeric Methods

Experience

Junior Fullstack Intern — Jan. 2026 – Present
Generis — *Toronto, ON*

- Maintain and update 21 live event websites in WordPress, ensuring uptime, content accuracy, and performance under tight production timelines.
- Developed full-stack web applications using React, PHP, and SQL, contributing to internal tooling and client-facing systems.
- Built custom WordPress plugins in PHP and automated recurring workflows in n8n, reducing manual staff overhead.
- Integrated and maintained Salesforce workflows to improve operational data consistency across teams.
- Provided IT support including workstation setup, hardware configuration, and peripheral troubleshooting; conducted equipment audits to assess lifecycle status and upgrade requirements.

Technical Team Member — Sept. 2025 – Present
UW Orbital — *Waterloo, ON*

- Design PCBs in Altium Designer for the team's in-house CubeSat, applying hardware design principles to real aerospace constraints.
- Apply Digital Signal Processing (DSP) concepts to optimize satellite radio communication, bridging software and hardware system design.

Student Researcher — Jan. 2025 – June 2025
University of Toronto — *Toronto, ON*

- Researched distance functions on multidimensional surfaces, drawing on topology, Riemannian geometry, and Lorentzian geometry.
- Developed a Python pipeline applying Riemannian geometry and UMAP to manifold learning, improving accuracy of Spotify music recommendation models.

Projects

ExoDiscover | *Python, Flask, React, Three.js, HTML/CSS, Scikit-learn* — Oct. 2025

- Architected an end-to-end pipeline from raw NASA dataset ingestion through AI inference to a live 3D React/Three.js interface, integrating two independently trained models into a unified web application.
- Integrated AI outputs into a React and Three.js web interface for 3D rendering of exoplanet data with interactive filters and parameter-based insights.
- Placed top 4 of 30 teams (150 participants) in a 6-person team during the 2025 NASA Space Apps Hackathon for developing ExoDiscover within 48 hours.

Manifold Learning for Spotify | *Python, UMAP, NumPY, Pandas, Matplotlib* — May 2025 – Present

- Built a machine learning pipeline applying UMAP to cluster Spotify track data.
- Created detailed visualizations with Matplotlib and packaged reproducible code for peer use.

Pulsar Classification System | *Python, Pandas, Matplotlib, Scikit-learn* — Sept. 2024 – Present

- Developed a predictive system to classify pulsar signals using Random Forest and SVM models.
- Cleaned and normalized large astronomy datasets, applying cross-validation and hyperparameter tuning.
- Evaluated performance with confusion matrices and ROC curves, ensuring accurate pulsar detection.